

1. Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a field of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence. These tasks may include understanding language, recognizing images, making decisions, solving problems, learning from data, and generating content.

AI systems are becoming increasingly common in everyday applications. Examples include recommendation systems on streaming platforms, voice assistants, navigation applications, spam filters, facial recognition systems, and conversational AI tools.

The main goal of AI is not necessarily to create machines that behave exactly like humans. Instead, AI can be used to develop systems that can perform specific tasks efficiently and make useful predictions or decisions based on available information.

1.1 Examples of AI Applications

AI can be applied to many different areas:

Healthcare:

AI can assist doctors by analyzing medical images, identifying patterns in patient data, and helping predict potential health risks.

Finance:

Financial institutions use AI for fraud detection, credit assessment, and analyzing large amounts of financial data.

Education:

AI can provide personalized learning recommendations, automatically generate practice questions, summarize educational materials, and assist students with revision.

Transportation:

AI can be used for route planning, traffic prediction, driver assistance systems, and autonomous vehicles.

Entertainment:

Streaming services use AI to recommend movies, music, and other content based on a user's previous activity.

1.2 AI and Automation

AI is closely related to automation, but the two concepts are not identical.

Traditional automation usually follows a fixed set of instructions. For example, a program might be instructed to automatically rename every file in a folder according to a specific naming convention.

AI systems can instead make decisions based on patterns or data. For example, an AI system could examine a collection of emails and determine whether a new email is likely to be spam.

Therefore, automation focuses primarily on following predefined rules, while AI can involve learning, prediction, reasoning, or interpretation.

2. Types of Artificial Intelligence

Artificial Intelligence can be classified in several different ways. One common classification is based on the capabilities of the AI system.

2.1 Narrow AI

Narrow AI, also called Weak AI, is designed to perform a specific task or a limited range of tasks.

Examples include:

- Image recognition systems
- Recommendation systems
- Voice assistants
- Spam detection
- Chess-playing programs
- AI writing assistants

A narrow AI system can perform its intended task very effectively, but it does not necessarily have general intelligence.

For example, an AI system designed to recognize cats in photographs may be extremely good at identifying cats. However, this does not mean that the same system can automatically solve mathematical problems, drive a car, or understand a completely unrelated task.

Most AI applications currently used in the real world fall under the category of narrow AI.

2.2 Artificial General Intelligence

Artificial General Intelligence (AGI) refers to a hypothetical type of AI that would have broad intellectual capabilities comparable to humans.

An AGI system would theoretically be able to learn and perform many different types of tasks rather than being limited to one specific application.

For example, a hypothetical AGI system might be able to:

- Learn a new programming language
- Understand a lecture
- Write an essay
- Solve mathematical problems
- Plan a trip
- Learn how to play a new game
- Analyze scientific information

AGI remains an area of research and discussion. It should not be confused with ordinary AI applications that are designed for specific purposes.

2.3 Artificial Superintelligence

Artificial Superintelligence (ASI) is a hypothetical form of AI that would surpass human intelligence across virtually all areas of intellectual ability.

ASI is primarily discussed as a theoretical concept rather than an existing technology.

Researchers and philosophers have discussed potential benefits and risks associated with highly advanced AI systems. These discussions include questions about safety, control, ethics, and how humans should interact with increasingly capable AI.

3. Machine Learning

Machine Learning (ML) is a major area of AI. Instead of programming every rule manually, machine learning allows computers to learn patterns from data.

A traditional program might follow instructions such as:

```
IF temperature is above 30°C  
THEN display "Hot"
```

A machine learning system may instead be provided with many examples containing temperatures and corresponding labels. The system can then learn a relationship between the input data and the desired output.

3.1 Training Data

Training data is the information used to teach a machine learning model.

For example, suppose we want to create a model that identifies whether an email is spam.

Our training data might contain:

```
Email 1 → Spam  
Email 2 → Not Spam
```

Email 3 → Spam
Email 4 → Not Spam

The model examines patterns in these examples and attempts to learn characteristics that distinguish spam from legitimate emails.

The quality of training data is extremely important. If the training data is incomplete, biased, incorrect, or poorly labeled, the resulting model may also perform poorly.

3.2 Features and Labels

A machine learning dataset commonly contains **features** and **labels**.

Features are pieces of information used as inputs to a model.

For example, when predicting whether a student will pass an examination, possible features could include:

- Hours studied
- Attendance percentage
- Number of assignments completed
- Previous examination scores

The label represents the outcome that the model is trying to predict.

For example:

Features:

Hours studied = 20

Attendance = 90%

Assignments = 8

Label:

Passed = Yes

3.3 Supervised Learning

Supervised learning uses training data where the correct answers are already known.

For example, a dataset might contain information about houses:

Size	Bedroom s	Locatio n	Price
900 sq ft	2	City	RM350,000
1200 sq ft	3	Suburb	RM420,000
1500 sq ft	4	City	RM600,000

The model can learn relationships between the features and house prices.

After training, the model can receive information about a new house and estimate its price.

3.4 Unsupervised Learning

Unsupervised learning works with data where there are no predefined labels.

The objective is often to discover patterns or groups within the data.

For example, a company may have information about thousands of customers but no predefined customer categories.

A clustering algorithm could identify groups of customers with similar behavior.

One possible result could be:

Group A → Frequent customers with high spending

Group B → Frequent customers with low spending

Group C → Infrequent customers with high spending

Group D → Infrequent customers with low spending

The algorithm discovers these groups based on patterns in the data.

4. Deep Learning and Neural Networks

Deep Learning is a subfield of machine learning that uses neural networks with multiple layers.

Neural networks are computational models inspired loosely by the structure of biological brains. They consist of interconnected units commonly called neurons.

A simplified neural network can be represented as:

Input Layer



Hidden Layer



Hidden Layer



Output Layer

Each layer processes information and passes the result to the next layer.

4.1 Input Layer

The input layer receives the original information.

For an image recognition system, the input could consist of numerical representations of pixels.

For a speech recognition system, the input could contain numerical representations of audio signals.

4.2 Hidden Layers

Hidden layers process information received from previous layers.

A deep learning model may contain many hidden layers. Different layers can learn different types of patterns.

For example, in an image recognition system:

Early layers → Simple edges and shapes

Middle layers → More complex patterns

Later layers → Objects or meaningful structures

This allows the network to build increasingly complex representations of the input.

4.3 Output Layer

The output layer produces the final result.

For example, an image classification model might produce:

Cat: 0.92

Dog: 0.06

Bird: 0.02

The model is indicating that it believes the image is most likely to contain a cat.

The numbers can be interpreted as confidence scores depending on the model and task.

5. Generative Artificial Intelligence

Generative AI refers to AI systems that can generate new content based on patterns learned from training data.

Generated content can include:

- Text
- Images

- Audio
- Video
- Computer code

Large Language Models (LLMs) are an important example of generative AI.

An LLM is trained on large amounts of text and learns statistical patterns in language. When given an input prompt, the model generates a response based on those learned patterns.

For example:

User:

Explain photosynthesis in simple terms.

AI:

Photosynthesis is the process plants use to convert light energy into chemical energy...

5.1 Large Language Models

Large Language Models are designed to process and generate natural language.

They can be used for tasks such as:

- Question answering
- Summarization
- Translation
- Text generation
- Classification
- Code generation
- Information extraction

However, an LLM does not necessarily function like a traditional database.

When a user asks a question, the model generates an answer based on its learned patterns and the information available in its context.

5.2 Prompts

A prompt is the input given to a generative AI system.

For example:

Explain machine learning in 100 words.

The quality and structure of a prompt can affect the resulting output.

A more specific prompt could be:

Explain supervised learning to a first-year

computer science student. Use one simple example and keep the explanation under 150 words.

This gives the model additional instructions about the target audience, content, and desired length.

6. AI APIs

An API, or Application Programming Interface, provides a standardized way for one software application to communicate with another software system.

For example, a Python application can send a request to an AI service through an API.

A simplified process looks like this:

Python Application



API Request



AI Service



AI Model



API Response



Python Application

The Python program does not necessarily need to run the AI model itself. Instead, it can send the required information to a remote service and receive the model's response.

6.1 API Requests

An API request may contain information such as:

- Authentication credentials
- Model name
- User input
- Configuration parameters

The service processes the request and sends a response back.

A response might contain generated text, classifications, numerical values, or other information depending on the API.

6.2 Why APIs Are Useful

APIs allow developers to integrate existing AI capabilities into their own applications without building and training an AI model from scratch.

For example, a student could create an AI Study Assistant that:

1. Accepts a student's question
2. Sends the question to an AI service
3. Receives the response
4. Displays the answer in a web application

This allows developers to focus on building useful applications around existing AI models.

7. Retrieval-Augmented Generation

Retrieval-Augmented Generation, commonly called RAG, is a technique that combines information retrieval with generative AI.

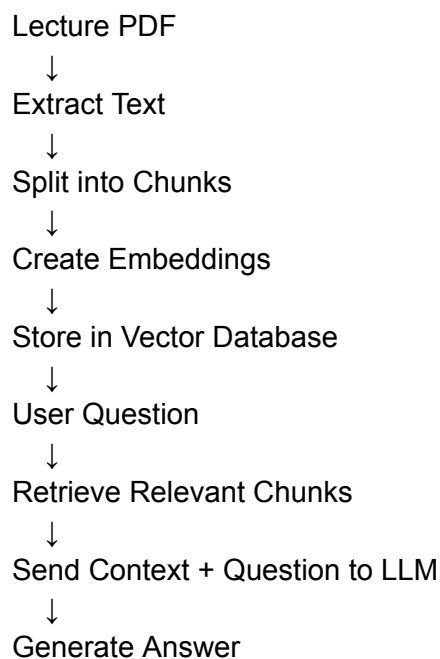
A normal language model may not have access to a user's private documents.

For example, imagine a student has a PDF containing lecture notes.

A basic AI system might not know the contents of that PDF.

A RAG system can instead retrieve relevant information from the document and provide that information to the language model.

A simplified RAG pipeline is:



7.1 Document Chunking

Large documents are often divided into smaller sections called chunks.

For example:

Lecture Notes



Chunk 1

Chunk 2

Chunk 3

Chunk 4

...

Chunk 50

When a user asks a question, the system searches for the chunks that are most relevant to the question.

7.2 Embeddings

Embeddings are numerical representations of information.

A piece of text can be converted into a vector containing many numerical values.

For example:

"Machine learning uses data"



[0.12, -0.43, 0.87, 0.21, ...]

Texts with similar meanings can have similar vector representations.

This allows a system to perform semantic searches rather than relying only on exact keyword matches.

7.3 Benefits of RAG

RAG can be useful when an application needs to answer questions based on a specific collection of documents.

Examples include:

- University lecture notes
- Company documentation
- Product manuals
- Research papers
- Legal documents
- Internal knowledge bases

Instead of asking the AI to rely only on its general knowledge, the application can provide relevant information from the selected documents.

8. Evaluating AI Systems

Building an AI system is only one part of an AI engineering project. Developers also need to evaluate whether the system works correctly.

Different AI tasks require different evaluation methods.

For classification, common metrics include:

- Accuracy
- Precision
- Recall
- F1-score

For generated text, evaluation can be more complicated because multiple answers may be reasonable.

For example, if a student asks:

What is supervised learning?

There may be many correct ways to explain the concept.

Therefore, evaluation may involve a combination of automated metrics and human assessment.

8.1 Accuracy

Accuracy measures the proportion of predictions that are correct.

For example, if a model makes 100 predictions and 90 are correct:

$$\begin{aligned}\text{Accuracy} &= 90 / 100 \\ &= 90\%\end{aligned}$$

Although accuracy is useful, it is not always sufficient. If the dataset is heavily imbalanced, a model can have high accuracy while performing poorly on an important class.

8.2 Testing AI Applications

AI applications should also be tested like other software systems.

For example, a document-processing application could have tests for:

Valid PDF
Invalid PDF
Empty PDF
Large PDF
Missing file
Incorrect file type

Automated tests can help developers detect problems when the application is changed.

9. Ethical Considerations in AI

AI systems can provide significant benefits, but they also introduce ethical and practical concerns.

9.1 Bias

AI models learn from data. If the training data contains biases, the model may reproduce or amplify those biases.

Developers should therefore consider the source, quality, and diversity of the data used to develop AI systems.

9.2 Privacy

AI applications may process sensitive or personal information.

For example, an AI Study Assistant could process:

- Student names
- Assignments
- Lecture notes
- Academic records
- Personal questions

Developers should carefully consider how this information is stored, processed, and protected.

9.3 Hallucinations

Generative AI systems can sometimes produce information that sounds convincing but is incorrect. This is commonly referred to as an AI hallucination.

For example, an AI system may provide a fictional source or incorrectly state a fact.

Therefore, AI-generated information should not automatically be treated as correct.

Applications that provide important information should include appropriate verification mechanisms.

10. Summary

Artificial Intelligence is a broad field that includes many different technologies and approaches.

The major concepts covered in this lecture include:

Artificial Intelligence:

The broader field of creating systems capable of tasks associated with intelligence.

Machine Learning:

A method where computers learn patterns from data.

Deep Learning:

A type of machine learning based on multi-layer neural networks.

Generative AI:

AI systems capable of generating new content such as text, images, audio, and code.

Large Language Models:

Models designed to understand and generate human language.

APIs:

Interfaces that allow software applications to communicate with external services.

RAG:

A technique that retrieves relevant external information and provides it to a generative AI model.

Embeddings:

Numerical representations of information that can be used for semantic similarity and retrieval.

AI Evaluation:

The process of measuring whether an AI system produces useful and reliable results.

AI Ethics:

The consideration of issues such as bias, privacy, safety, and reliability.

Understanding these concepts provides a foundation for developing practical AI applications. Modern AI engineering often involves combining multiple technologies rather than building a single model from scratch. A typical application may use Python for its backend logic, an API

to access an AI model, embeddings and a vector database for document retrieval, and a web framework to provide an interface for users.